

RAFELA NARANLELE

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PROFESSIONAL PROFILE

Marine and coastal ecology graduate with field-based research experience in mangrove, seagrass, coral reef, limnological, and coastal pollution assessments. Trained in standardized ecological survey methods, physicochemical water quality measurements, plankton and benthic sampling, and GIS-based spatial analysis. Experienced in applying ecosystem-based and land sea interaction approaches to evaluate environmental conditions and support coastal management strategies. Seeking to advance research in marine ecosystem dynamics and sustainable coastal systems.

EDUCATION

Bachelor of Fisheries Science (S.Pi.) – Aquatic Resource Management

Faculty of Fisheries and Marine Science, Pattimura University

2017–2021 | CGPA: 3.59/4.00

Undergraduate Thesis:

Coastal Spatial Management in Atubul Da Village, Tanimbar Islands Regency

- Conducted comprehensive ecological assessments across 4 coastal sampling sites encompassing mangrove (± 5.40 ha), seagrass (± 380.61 ha), and coral reef ecosystems (± 409.59 ha).
- Implemented belt transect (10×10 m; 5×5 m; 1×1 m), systematic line transect, and 50 m Line Intercept Transect (LIT) methodologies.
- Measured key ecological indicators including vegetation density, species composition, and benthic cover.
- Performed spatial mapping and coastal zoning analysis using ArcGIS.
- Developed SWOT-based coastal management strategy integrating ecological and land sea interaction indicators

RESEARCH EXPERIENCE

Coastal & Marine Ecosystem Assessment (June 2018 - January 2021)

Research Team Leader & Research Assistant

- Led a field research team in coastal ecosystem assessments, coordinating sampling design and data collection activities.
- Conducted mangrove vegetation structure analysis using multi-scale quadrat sampling
- Evaluated seagrass density and species composition using systematic transect-quadrat design
- Contributed to biodiversity documentation and ecosystem health evaluation under academic supervision.

Coastal Pollution & Microbial Assessment (Inner Ambon Bay) (December 2019)

Research Assistant

- Collected seawater and gastropod samples in densely populated coastal zones to assess anthropogenic contamination.
- Assisted in laboratory identification of *Salmonella* spp. as microbial indicators of environmental pollution.
- Contributed to nearshore marine environmental quality assessment and data documentation.

River Limnology & Water Quality Assessment (Wailela River) (December 2019)

Research Team Leader & Research Assistant

- Led field-based limnological assessments across upstream, midstream, and estuarine zones, coordinating sampling design and team activities.
- Measured key physicochemical parameters including temperature, pH, dissolved oxygen (DO), current velocity, and water transparency.
- Collected plankton, benthos, and periphyton samples to evaluate freshwater productivity and ecological condition.
- Contributed to integrated water quality assessment under academic supervision.

Plankton Ecology Survey (December 2019 - March 2021)

Research Team Leader & Research Assistant

- Led field-based plankton surveys across 4 marine environments (estuary, aquaculture zone, mangrove area, and inner bay), coordinating sampling logistics and data documentation.
- Conducted vertical and horizontal plankton net sampling to assess phytoplankton and zooplankton distribution.
- Assisted in laboratory identification and abundance analysis to support ecological assessment under academic supervision.

Benthic Bioindicator Study (Seith Village, Ambon) (June 2021)

Research Assistant

- Assisted in field assessment of benthic macrofauna across mangrove, seagrass, and sedimentary substrates.
- Collected and documented abundance data of gastropods and bivalves as environmental bioindicators.
- Contributed to benthic community structure analysis to support ecological condition evaluation.

RESTORATION VOLUNTEER

Poka–Rumah Tiga, Inner Ambon Bay (May 2019)

- Conducted mangrove restoration activities through the planting of approximately 200 seedlings of *Rhizophora mucronata* and *Rhizophora apiculata* as part of coastal ecosystem rehabilitation efforts.

Negeri Naku, Estuarine Area, Ambon (March 2021)

- Participated in mangrove reforestation by planting 100 seedlings of *Rhizophora mucronata* in a river estuary zone to support shoreline stabilization and habitat recovery.

CERTIFICATION

Assessor of Seagrass

Indonesian Professional Certification Authority (BNSP), 2022

- Certified competent in seagrass bed condition assessment and evaluation of ecological indicators

TECHNICAL & ANALYTICAL SKILLS

- Marine ecological survey methods (transect–quadrat, LIT)
- Plankton net sampling (vertical & horizontal)
- Benthic and vegetation community analysis
- Physicochemical water quality measurement (DO, pH, temperature, salinity)
- ArcGIS spatial analysis and coastal mapping
- SWOT-based environmental management framework
- Scientific data interpretation and academic reporting

AWARDS & HONORS

- Global Goals Youth Initiator 2022
- 1st Runner Up – SDGs Video Campaign 2022
- Honourable Delegate – Asia Pacific Youth SDGs Summit, 2022 (Virtual)
- Honourable Delegate – Asia Global Students Conference & Exchange, 2022 (Virtual)